THERMTEC INDUSTRIAL INSTRUMENTS

TH200 Smart Temperature Transmitter

4–20 mA / Modbus RTU / HART

Installation, Configuration & Communication Manual

Document No. TH200-MAN-003 | Revision 4.1 | June 2026

■ READ THIS MANUAL BEFORE INSTALLATION. Failure to follow these instructions may result in equipment damage, process upset, or personal injury.

1. Product Overview

The TH200 is a two-wire, loop-powered smart temperature transmitter designed for demanding industrial process control applications. It accepts inputs from Type J, K, T, E, R, S, B thermocouples and 100 Ω / 1000 Ω Pt RTDs. The 4–20 mA analogue output is supplemented by a full Modbus RTU serial interface on the integrated RS-485 port, enabling digital configuration and continuous register-mapped readback without breaking the analogue loop.

Key features:

Feature	Specification
Measurement input	TC: J, K, T, E, R, S, B | RTD: Pt100 / Pt1000 (2/3/4-wire)
Analogue output	4–20 mA, max load 600 Ω at 24 V DC
HART revision	HART 7 (optional factory option)
Digital interface	Modbus RTU over RS-485 (2-wire, half-duplex)
Slave address range	1–247 (factory default: 1)
Baud rates supported	1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200
Default baud rate	9600 bps
Data format	8 data bits, No parity, 1 stop bit (8N1)
Process temperature	–40 °C to +120 °C (sensor dependent)
Ambient temperature	–40 °C to +85 °C
Supply voltage	12–36 V DC (two-wire loop)
Protection class	IP66 / NEMA 4X
Certifications	ATEX Zone 1/2, IECEx, SIL 2 (IEC 61508)
Housing material	316L stainless steel
Process connection	1/2" NPT or G1/2 thread

Ordering code: TH200 – [Input] – [Output] – [Housing] – [Options]

2. Safety Instructions

2.1 General

Installations must comply with local regulations, IEC 60079 (hazardous areas), and applicable electrical codes. Only qualified personnel with knowledge of process instrumentation and relevant safety standards should install, operate, or maintain this equipment.

2.2 Electrical Safety

Maximum loop supply: 36 V DC. Do not exceed this value. The device contains no user-serviceable parts. Return to factory or authorised service centre for repair.

2.3 Process Safety Constraints

The following safety limits are enforced by the device firmware and MUST be respected in any configuration:

Parameter	Minimum	Maximum	Consequence if violated
Loop current	3.6 mA	21.0 mA	Open/short circuit detection
Process temperature	–40 °C	+120 °C	Overrange error (22 mA burnout)
Supply voltage	12 V DC	36 V DC	Under/overvoltage protection
Slave address	1	247	Address 0 is broadcast only
Baud rate	1200 bps	115200 bps	Communication fault

3. Wiring and Installation

3.1 Terminal Identification

Terminal	Label	Description
1	+	Loop supply positive (4–20 mA, +V)
2	–	Loop supply negative / signal return
3	A	RS-485 A (non-inverting, +)
4	B	RS-485 B (inverting, –)
5	⏚	Functional earth (shield)
6/7	SEN+/SEN–	RTD or thermocouple sensor input
8	SEN_REF	RTD third wire (3-wire RTD only)

3.2 RS-485 Network Wiring

Use shielded twisted-pair cable (e.g. Belden 9841 or equivalent, 120 Ω characteristic impedance). Terminate the bus at both physical ends with 120 Ω resistors. Maximum bus length: 1200 m at 9600 bps. Each TH200 presents a unit load of 1/8 on the RS-485 bus, allowing up to 256 devices with standard transceivers.

4. Modbus RTU Communication

4.1 Protocol Parameters

Parameter	Default	Configurable via
Slave address	1	Local keypad or Modbus FC06 (register 40201)
Baud rate	9600	Local keypad or Modbus FC06 (register 40202)
Parity	None	Local keypad or Modbus FC06 (register 40203)
Stop bits	1	Fixed
Byte order	Big-endian (MSB first)	Fixed
Turnaround delay	≥ 5 ms	Fixed
Broadcast support	Yes (address 0, write-only)	

4.2 Supported Function Codes

Code	Name	Used for
FC01	Read Coils	Read status bits (alarm, diagnostics)
FC03	Read Holding Registers	Read process values and configuration
FC04	Read Input Registers	Read live measurement registers (read-only)
FC06	Write Single Register	Write a single configuration register
FC16	Write Multiple Registers	Write a block of configuration registers

4.3 Register Map

Register (40xxx)	Address (hex)	R/W	Data type	Description	Units / Notes
40001	0000	R	INT16 × 10	Primary process value (temperature)	°C × 10, e.g. 256 = 25.6 °C
40002	0001	R	INT16 × 10	Sensor 2 / ambient temperature	°C × 10
40003	0002	R	UINT16	Device status word	See Table 4-4
40004	0003	R	UINT16	Loop current (mA × 100)	e.g. 1200 = 12.00 mA
40010	0009	R	UINT32 BE	Primary value as IEEE 754 float (high)	Registers 40010–40011
40011	000A	R	UINT32 BE	Primary value as IEEE 754 float (low)	Combined with 40010
40101	0064	R	UINT16	Firmware version (BCD)	e.g. 0x0401 = v4.1
40102	0065	R	UINT16	Hardware revision	e.g. 0x0200 = rev 2.0
40103	0066	R	UINT32 BE	Serial number (high word)	Registers 40103–40104
40104	0067	R	UINT32 BE	Serial number (low word)	Combined with 40103
40201	00C8	R/W	UINT16	Modbus slave address	Range 1–247, default 1

Register (40xxx)	Address (hex)	R/W	Data type	Description	Units / Notes
40202	00C9	R/W	UINT16	Baud rate code	0=1200,1=2400,2=4800,3=9600,4=1920
40203	00CA	R/W	UINT16	Parity / stop bits	0=8N1, 1=8E1, 2=8O1
40210	00D1	R/W	INT16 × 10	Range low (4 mA setpoint)	°C × 10, default –400 (–40.0 °C)
40211	00D2	R/W	INT16 × 10	Range high (20 mA setpoint)	°C × 10, default 1200 (+120.0 °C)
40212	00D3	R/W	UINT16	Burnout direction	0=upscale (21.0 mA), 1=downscale (3.6
40213	00D4	R/W	UINT16	Damping time constant	Seconds × 10, range 0–600
40220	00DB	R/W	UINT16	Sensor type	0=K-TC,1=J-TC,2=T-TC,3=E-TC,4=Pt10
40221	00DC	R/W	UINT16	RTD wiring mode	2=2-wire,3=3-wire,4=4-wire
40230	00E5	R/W	INT16 × 100	User calibration offset	°C × 100, default 0
40231	00E6	R/W	INT16 × 1000	User calibration gain	× 1000, default 1000
40240	00EF	R/W	UINT16	High temperature alarm setpoint	°C × 10
40241	00F0	R/W	UINT16	Low temperature alarm setpoint	°C × 10
40242	00F1	R/W	UINT16	Alarm hysteresis	°C × 10, default 20 (2.0 °C)
40300	012B	R/W	UINT16	Factory reset	Write 0xA5A5 to restore defaults

4.4 Status Word Bit Map (Register 40003)

Bit	Name	Description
0	PV_VALID	1 = process value is valid
1	SENSOR_FAULT	1 = sensor open/short circuit detected
2	OVERRANGE	1 = input above high range limit
3	UNDERRANGE	1 = input below low range limit
4	HIGH_ALARM	1 = high temperature alarm active
5	LOW_ALARM	1 = low temperature alarm active
6	LOOP_FAULT	1 = loop current < 3.5 mA or > 21.5 mA
7	NV_WRITE_ERROR	1 = non-volatile memory write failure
8	CAL_ACTIVE	1 = calibration mode active
9–15	RESERVED	Always 0

5. Device Configuration

5.1 Setting Slave Address via Modbus

To change the slave address, write the new address (1–247) to register 40201 using FC06. The change takes effect immediately and is stored in non-volatile memory. Broadcast address 0 cannot be set as the slave address.

Example FC06 PDU (set slave address to 3):

```
TX: 01 06 00 C8 00 03 [CRC] → RX: 01 06 00 C8 00 03 [CRC]
```

5.2 Setting Baud Rate

Write a baud rate code (0–7) to register 40202. The device will apply the new baud rate after a 500 ms delay. Ensure the host adapts immediately. Both ends must match.

Example: set 19200 bps (code 4):

```
TX: 01 06 00 C9 00 04 [CRC]
```

5.3 Sensor Configuration

Write the sensor type code to register 40220. For RTD sensors, also write the wiring mode to register 40221. The device will automatically recalculate linearisation tables.

5.4 Scaling the 4–20 mA Output

Registers 40210 (low range, 4 mA) and 40211 (high range, 20 mA) define the analogue output scaling. Values are in °C × 10. For example, to set range –40 to +120 °C:

```
40210 = 0xFF70 (–400 = –40.0 °C) 40211 = 0x04B0 (1200 = +120.0 °C)
```

6. OPC-UA Companion Interface (optional)

The TH200-UA factory option adds an embedded OPC-UA server (IEC 62541). The server is accessible at the default endpoint:

```
opc.tcp://<device-ip>:4840/TH200
```

6.1 NodeId Map

NodeId (ns=2)	Browse Name	Data Type	Access	Description
i=1001	ProcessValue	Float	RO	Temperature in °C
i=1002	LoopCurrent	Float	RO	4–20 mA loop current
i=1003	DeviceStatus	UInt16	RO	Status word (see §4.4)
i=2001	SlaveAddress	UInt16	RW	Modbus slave address
i=2002	BaudRate	UInt32	RW	Baud rate in bps
i=2010	RangeLow	Float	RW	4 mA setpoint (°C)
i=2011	RangeHigh	Float	RW	20 mA setpoint (°C)
i=2020	SensorType	String	RW	e.g. 'Pt100', 'K-TC'
i=9001	Manufacturer	String	RO	'THERMTEC INDUSTRIAL'
i=9002	ModelNumber	String	RO	'TH200'
i=9003	SerialNumber	String	RO	Device serial number
i=9004	FirmwareVersion	String	RO	e.g. '4.1.0'

7. Troubleshooting

Symptom	Likely cause	Remedy
No response to Modbus queries	Wrong slave address or baud rate	Check address/baud via keypad; use address 0 broadcast to d
Output stuck at 21 mA	Sensor open circuit (burnout up)	Check sensor wiring; verify register 40003 bit 1
Output stuck at 3.6 mA	Sensor short (burnout down)	Check sensor wiring; verify register 40003 bit 1
CRC errors	Termination missing	Add 120 Ω at both bus ends
OVERRANGE / UNDERRANGE alarm	Range limits too narrow	Adjust registers 40210/40211
OPC-UA connection refused	Non-UA variant ordered	Confirm order code includes -UA suffix
Calibration drift	Ambient temperature extremes	Apply user offset via register 40230

8. Complete Electrical Specifications

Parameter	Value	Condition
Supply voltage	12–36 V DC	Two-wire loop
Quiescent current	< 3.6 mA	4 mA output, 24 V supply
Output current range	3.6–21.0 mA	Including NAMUR NE43 levels
Load resistance	0–600 Ω	At 24 V DC supply
Accuracy	±0.1 °C	Pt100, 0–100 °C, 23 °C ambient
Long-term stability	< 0.1 °C / year	Typical
Response time (T63)	≤ 5 s	Damping = 0
EMC immunity	EN 61326-1, Class A	—
Safety integrity	SIL 2 (IEC 61508)	Hardware fault tolerance 0
MTBF	> 50 years	Siemens SN 29500, 40 °C
RS-485 unit load	1/8	Up to 256 devices
RS-485 max. bus length	1200 m	At 9600 bps, 120 Ω terminated
Power consumption	< 0.5 W	Typical at 24 V, 12 mA

© 2026 THERMTEC Industrial Instruments GmbH. All rights reserved. TH200-MAN-003 Rev 4.1.

Specifications subject to change without notice. Visit www.thermtec.example.com for the latest documentation.